

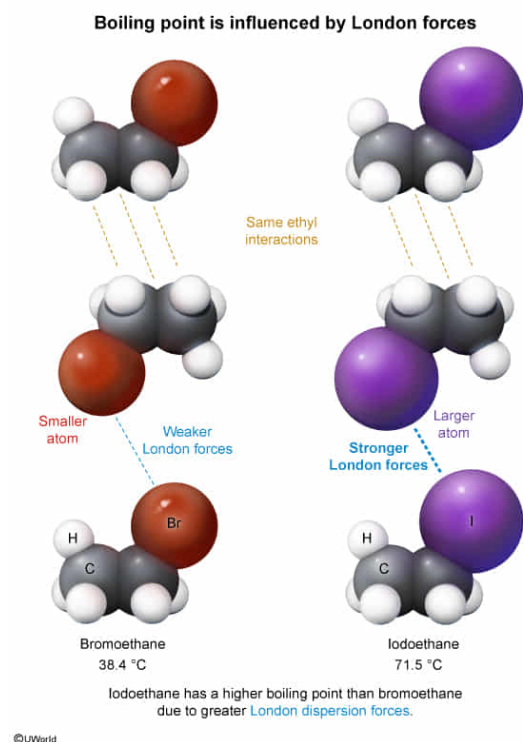
The boiling point of $\text{CH}_3\text{CH}_2\text{Br}$ (bromoethane) is $38.4\text{ }^\circ\text{C}$ whereas the boiling point of $\text{CH}_3\text{CH}_2\text{I}$ (iodoethane) is $71.5\text{ }^\circ\text{C}$. Which of the following best explains the difference between the boiling points of the two alkyl halides?

- ☐ A. The C–Br bond in bromoethane is less reactive than the C–I bond in iodoethane. [15%]
- ☐ B. The bromine atom in bromoethane is more polarizable than the iodine atom in iodoethane. [12%]
- ☐ C. The C–Br bond in bromoethane is more polar than the C–I bond in iodoethane. [22%]
- ✓ ☒ D. Iodoethane has greater London dispersion forces than bromoethane. [48%]

Correct

48% answered correctly

Explanation:



The boiling points of compounds are significantly influenced by **intermolecular forces**, which cause attractions between molecules. Structures with **weaker intermolecular forces** have **lower boiling points** because less energy is required to overcome these attractions between molecules during the **phase change** from a liquid to a gas.

London dispersion forces are intermolecular attractions between **instantaneous dipoles** induced by momentary distortions in the distribution of the electron "cloud" around atoms. As such, London dispersion forces are greater in larger molecules, which have a larger electron cloud and thus are **more polarizable**.

London dispersion forces are among the weakest intermolecular forces (as an individual interaction). However, the combined effect of numerous dispersion force interactions often

make London dispersion forces a very strong net influence on physical properties, especially in larger molecules.

Comparing bromoethane and iodoethane, both molecules have an equivalent, nonpolar CH_3CH_2 chain, but the iodine substituent of iodoethane is **larger** and **more polarizable** than the bromine substituent of bromoethane. As such, iodoethane experiences greater London dispersion forces and has a higher boiling point than bromoethane.

(Choice A) Boiling is a *physical process*, not a chemical reaction. As such, the reactivity of the bonds is irrelevant. However, if a reaction were being considered, the species responding at a lower temperature would be *more reactive*, not less reactive.

(Choice B) Bromine atoms are smaller and *less* polarizable than iodine atoms. Because iodine atoms polarize more easily, they contribute greater London forces than bromine atoms when present as substituents in molecules.

(Choice C) Although the C–Br bond is slightly more polar than the C–I bond, the dipole is negligible, and the lower boiling point of bromoethane shows that any effect from C–Br dipole-dipole interactions does not compete with the more significant London forces in iodoethane.

Educational objective:

Structures with stronger intermolecular forces have higher boiling points because more energy is required to overcome the attractions between molecules. Larger molecules tend to have higher boiling points because the increased size causes greater London dispersion forces.

Time spent:0 sec

Subject:
General Chemistry

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System:
Interactions of Chemical Substances